

# AI Portfolio Assistant and AI Appointment Booking Assistant using n8n, Google Sheets, Open Router, and Eleven Labs

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**Topic: No-Code Clinical NLP Workflow  
using Eleven Labs and n8n**

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**Course: AIGC 5600 – Generative AI and Prompt Eng**

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## **Class Activity 1: AI Portfolio Assistant using n8n, Google Sheets, and ElevenLabs**

### **Introduction**

During this class activity, I built an AI-powered portfolio assistant by integrating n8n, Google Sheets, Open Router, and Eleven Labs. The objective was to create a smart assistant that could answer questions about investment holdings using real-time data stored in Google Sheets. The workflow automatically retrieves portfolio information, processes the user's query with an AI model, and provides a clear, conversational response. This activity helped me understand how no-code AI tools can work together to automate data retrieval, improve user interaction, and build practical AI-powered solutions without extensive programming.

### **Objective**

The main goal of this class activity was to learn how AI Agents can interact with external tools through workflow automation. I explored how n8n, Google Sheets, OpenRouter, and ElevenLabs work together to process user requests and generate intelligent responses. This activity also helped me understand tool calling, voice-based AI interactions, and the practical use of no-code AI platforms for building real-world applications.

### **Workflow Explanation**

The workflow begins when the user asks a question about their investment portfolio through the AI assistant. The request is sent to the n8n workflow, where the AI Agent analyzes the query and determines whether it needs to retrieve information from Google Sheets. If portfolio data is required, the Google Sheets tool is automatically called to fetch the relevant details. The retrieved data is then processed by the OpenRouter language model, which generates a clear, accurate, and conversational response. Finally, the answer is returned to the user through the n8n interface or the ElevenLabs voice assistant.

### **Class Activities**

- Created the workflow in n8n.
- Connected Google Sheets.
- Connected Open Router Chat Model.
- Tested portfolio questions.
- Integrated Eleven Labs voice assistant.
- Verified successful workflow execution.

### **Results**

The AI assistant successfully answered different questions about the investment portfolio by retrieving accurate information from Google Sheets. It was able to identify portfolio holdings, provide relevant details, and generate natural, easy-to-understand responses. Throughout the testing process, the workflow performed reliably and demonstrated effective communication between the AI Agent, Google Sheets, OpenRouter, and n8n. This

activity showed how no-code AI tools can be integrated to automate data retrieval and deliver intelligent responses, creating a practical and user-friendly AI assistant.

## Reflection

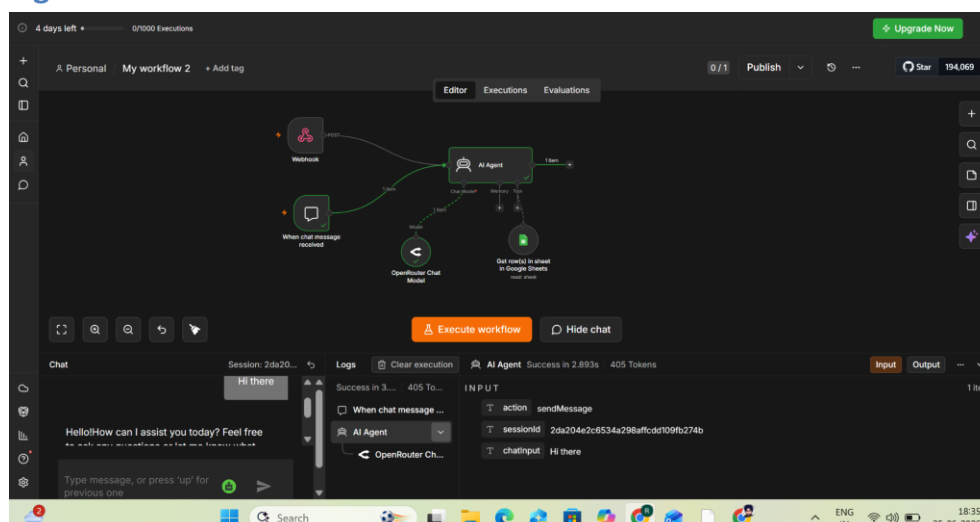
This class activity gave me valuable hands-on experience with AI workflow automation and helped me understand how different no-code AI tools can work together to build intelligent applications. While creating the workflow, I learned how AI Agents communicate with external tools such as Google Sheets to retrieve real-time information and how the Open Router language model generates meaningful responses based on the retrieved data. I also encountered a few workflow and configuration issues during testing, which improved my troubleshooting and problem-solving skills. Resolving these challenges gave me a better understanding of debugging AI workflows and integrating multiple services successfully. Overall, this activity increased my confidence in designing practical AI solutions and showed me how no-code platforms can simplify the development of smart, interactive applications for real-world use.

## Conclusion

This class activity successfully demonstrated how no-code AI tools can be integrated to build an intelligent portfolio assistant. By combining n8n, Google Sheets, OpenRouter, and ElevenLabs, I was able to automate the process of retrieving portfolio information and generating conversational responses. Throughout the activity, I gained practical experience in AI workflow automation, tool calling, and data integration. Overall, this project strengthened my understanding of no-code AI development and showed how different AI services can work together to create practical, efficient, and user-friendly solutions for real-world applications.

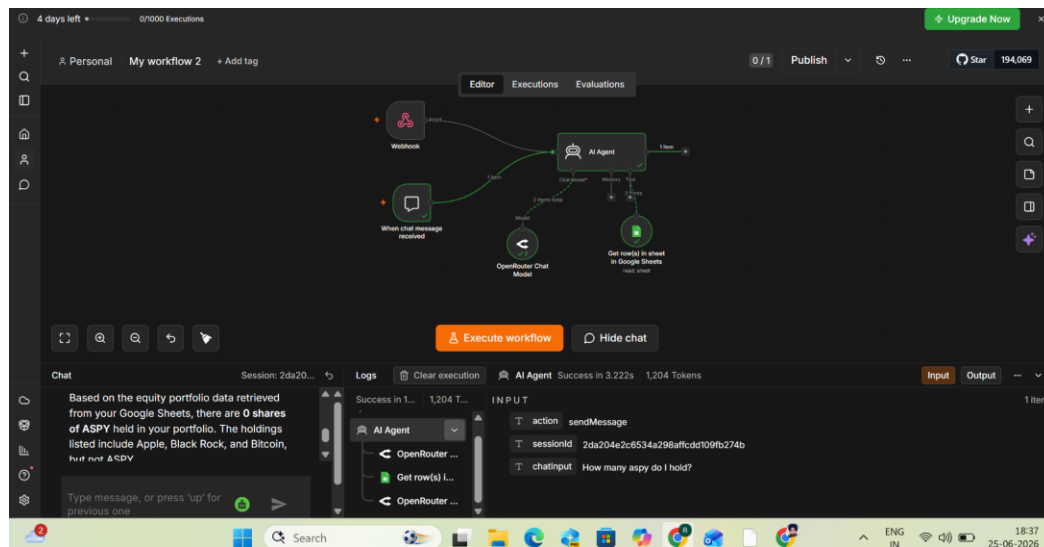
## Screenshots

Figure 1. Initial Portfolio Workflow



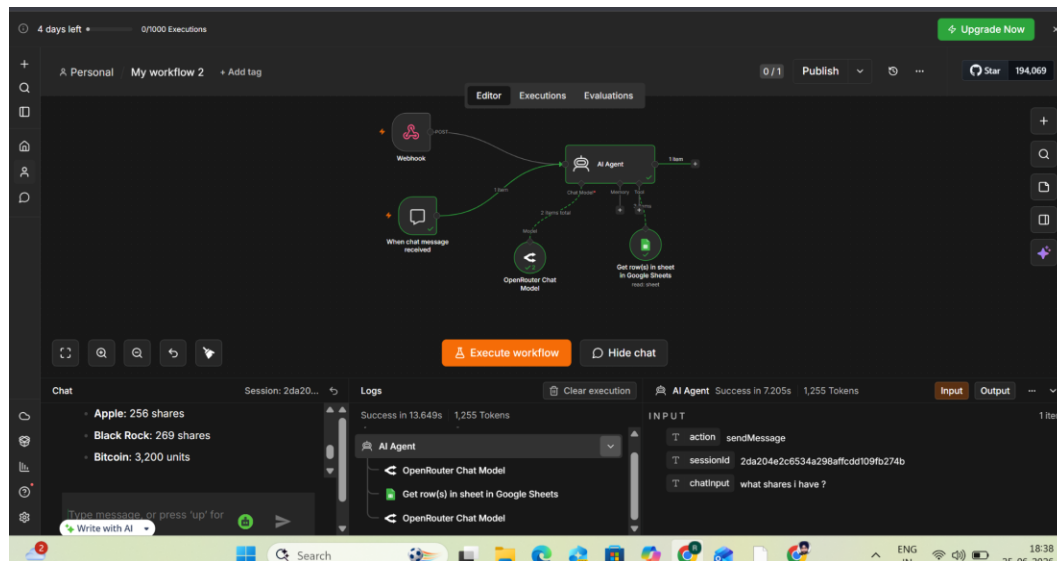
Workflow creation.

Figure 2. Testing ASPY Portfolio Query



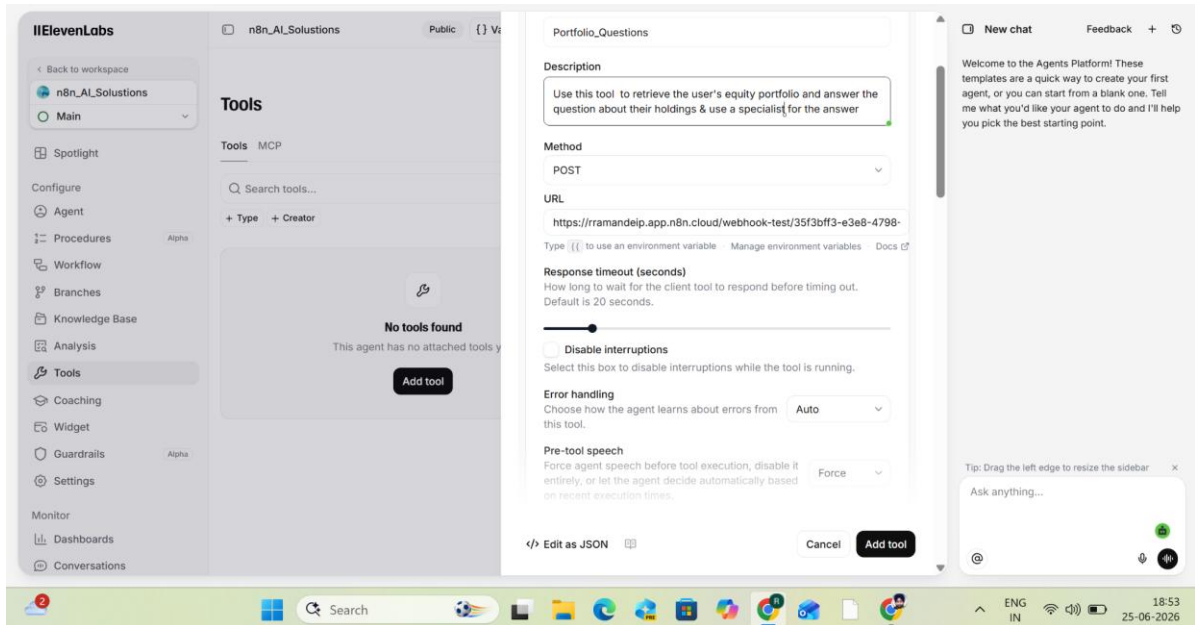
Testing ASPY question.

Figure 3. Portfolio Holdings Response



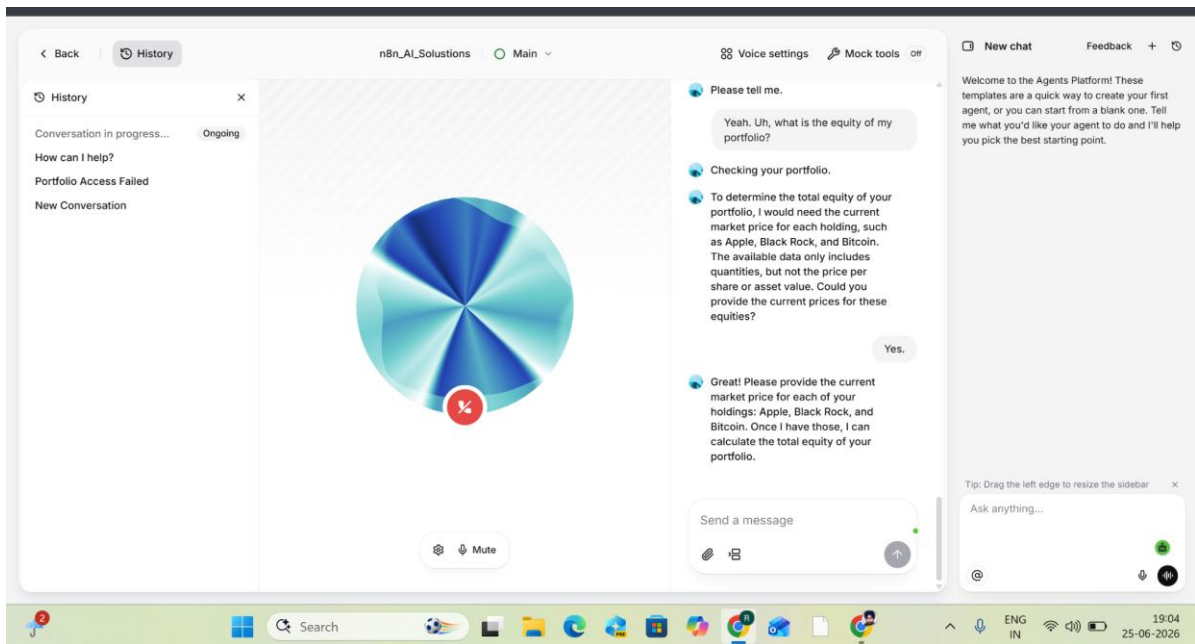
Portfolio holdings output.

Figure 4. ElevenLabs Tool Configuration



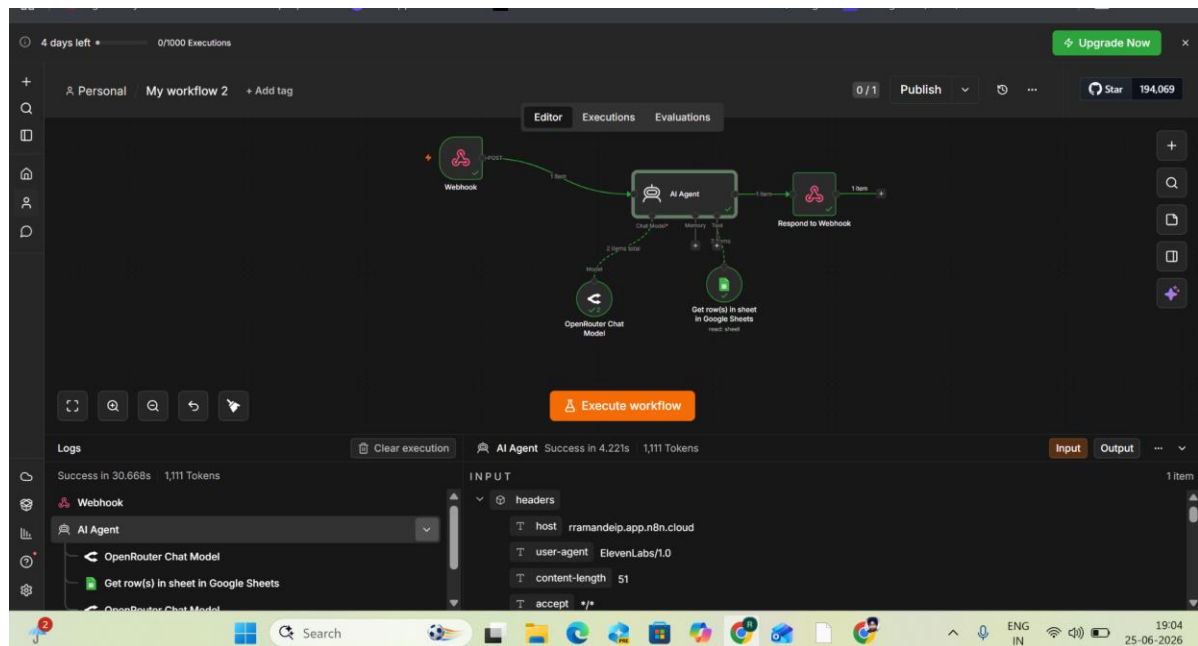
ElevenLabs tool configuration.

Figure 5. Voice Interaction



Voice interaction.

Figure 6. Successful Workflow Execution



Successful execution.

## Activity 2: AI Appointment Booking Assistant using Eleven Labs and n8n

### 1. Introduction

This lab demonstrates how a voice-enabled AI assistant can automate the appointment booking process by combining ElevenLabs, n8n, and the OpenRouter language model. The assistant listens to the user's voice request, understands important details such as the person's name, purpose of the meeting, preferred date, time, and email address, and sends this information to an n8n workflow through a webhook. The AI Agent processes the collected details and generates a professional appointment request email in a clear and well-structured format. Throughout this activity, I gained practical experience in building a no-code AI workflow, integrating multiple AI services, and testing end-to-end automation. The project highlighted how voice AI and workflow automation can improve efficiency, reduce manual effort, and provide a more natural and user-friendly experience.

## 2. Workflow Overview

The workflow begins when the user interacts with the ElevenLabs voice assistant by speaking naturally about an appointment request. The assistant captures the user's voice, identifies important details such as the person's name, meeting purpose, preferred date, time, and email address, and sends this information to an n8n workflow through a webhook. Inside n8n, the AI Agent receives the structured data and processes it using the OpenRouter language model along with the predefined system prompt. Based on the provided information, the AI generates a professional appointment request email in a clear and organized format. Finally, the completed email is returned through the Respond to Webhook node and displayed to the user, creating a smooth, fully automated, no-code workflow.

## 3. ElevenLabs Tool Configuration

In ElevenLabs, I created a custom tool called Book Appointment to handle voice-based appointment requests. The tool was configured with required parameters, including the user's name, purpose of the meeting, preferred date, preferred time, and email address. Marking these fields as required ensured that the assistant collected all essential information before sending the request to the n8n workflow. I also configured the webhook URL so the captured details could be transferred automatically for further processing. This setup enabled smooth communication between Eleven Labs and n8n, allowing the AI assistant to accurately understand the user's request and generate a professional appointment email without requiring any manual input.

## 4. n8n Workflow Configuration

The n8n workflow is built using four main nodes that work together to automate the appointment request process. The Webhook node acts as the entry point by receiving the appointment details sent from the Eleven Labs voice assistant in JSON format. The information is then passed to the AI Agent, which organizes the collected data into a structured prompt using dynamic expressions. Next, the Open Router Chat Model processes the prompt and generates a professional appointment request email based on the user's information. Finally, the Respond to Webhook node sends the generated email back to Eleven Labs, allowing the user to receive the response instantly through a smooth and fully automated workflow.

## 5. Testing and Results

Several test executions were performed to verify that the workflow functioned correctly from start to finish. During testing, the AI assistant successfully collected the user's appointment details, processed the information through the n8n workflow, and generated a professional appointment request email. The email included a clear subject line, an appropriate greeting, the meeting purpose, preferred date and time, contact information,

and a professional closing. The successful communication between Eleven Labs, n8n, and the Open Router language model demonstrated that the workflow was reliable, efficient, and capable of automating the appointment request process without requiring any manual coding.

## 6. Reflection

During the development of this workflow, I encountered several challenges, particularly with prompt mapping, webhook configuration, and correctly passing data between ElevenLabs and n8n. Initially, some of the required fields were not being processed correctly, which resulted in workflow errors. By carefully reviewing the execution logs, testing different JSON expressions, and adjusting the AI Agent configuration, I was able to resolve these issues successfully. This experience improved my understanding of webhook communication, prompt engineering, workflow debugging, and AI automation. It also strengthened my confidence in troubleshooting integration problems and building reliable no-code AI applications.

## 7. Conclusion

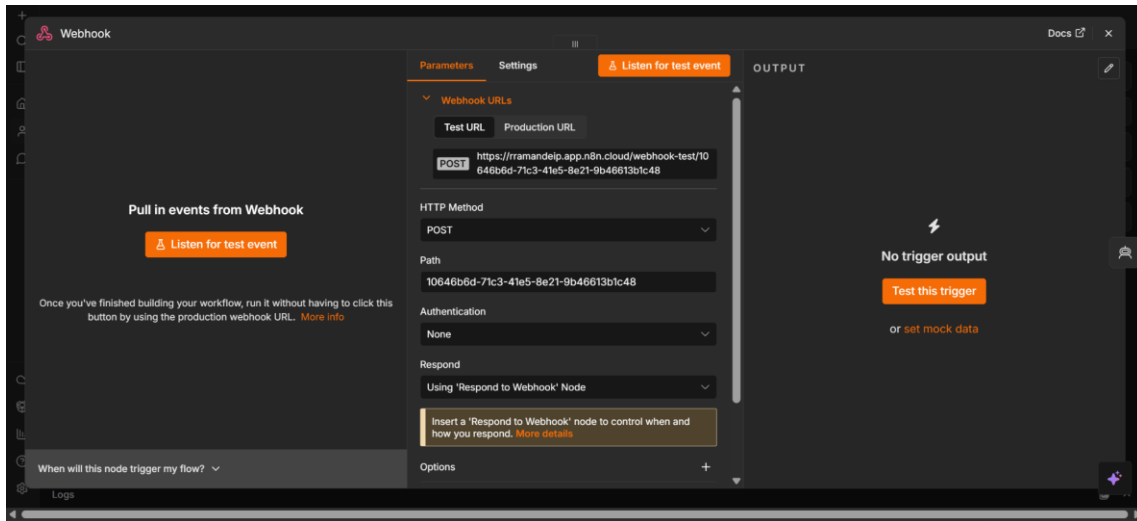
This project successfully demonstrated how different no-code AI tools can be integrated to create a complete voice-enabled appointment booking workflow. By combining Eleven Labs, n8n, and the Open Router language model, I was able to automate the process of collecting appointment details and generating a professional appointment request email. Throughout this lab, I gained valuable hands-on experience with AI workflow automation, webhook integration, prompt engineering, and troubleshooting real-world configuration issues. Working through these challenges strengthened my problem-solving skills and improved my confidence in building reliable AI workflows. Overall, this activity enhanced my understanding of no-code AI development and demonstrated how intelligent automation can simplify repetitive tasks, improve efficiency, and deliver practical solutions for real-world applications.

## 8. Screenshots and Explanation

### Figure 1. n8n Webhook Configuration

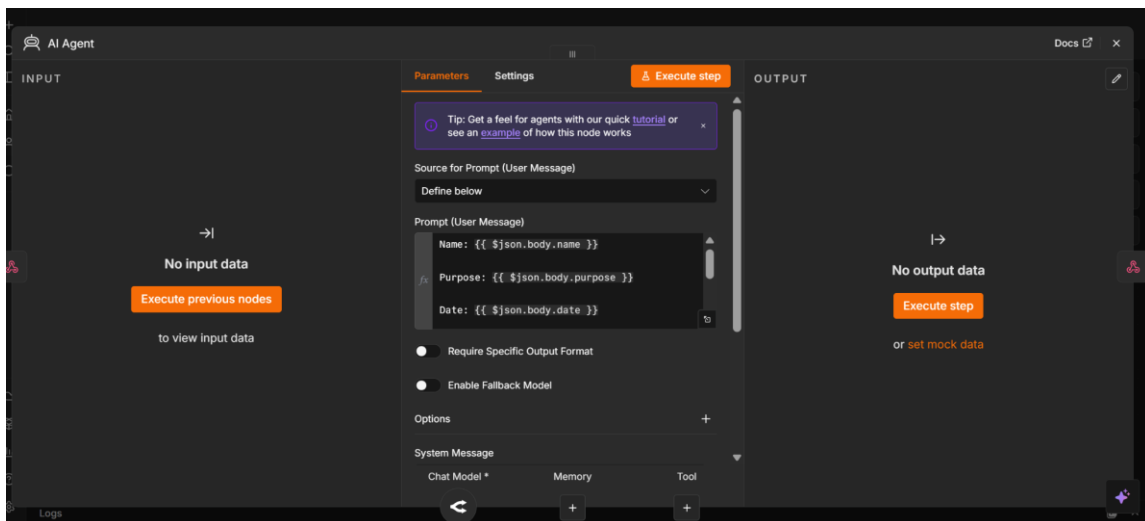
Webhook configuration in n8n showing the POST endpoint and webhook settings.





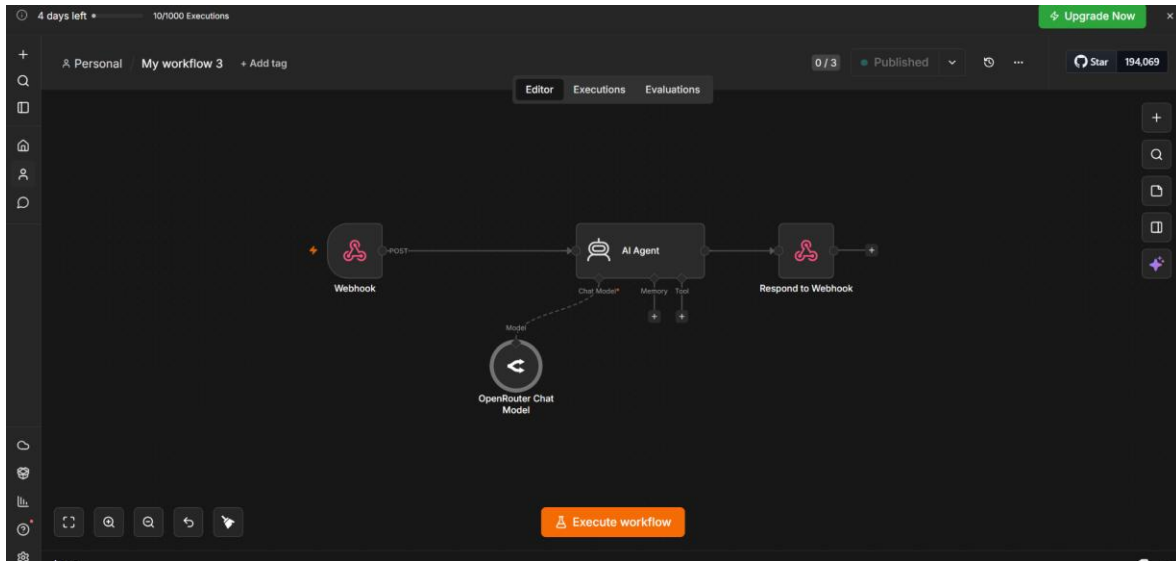
**Figure 2. AI Agent Prompt Configuration**

AI Agent configuration using dynamic prompt variables.



**Figure 3. Complete n8n Workflow**

Complete n8n workflow connecting Webhook, AI Agent, OpenRouter Chat Model and Respond to Webhook.



**Figure 4. ElevenLabs Tool Configuration**

ElevenLabs custom tool configuration with webhook URL.

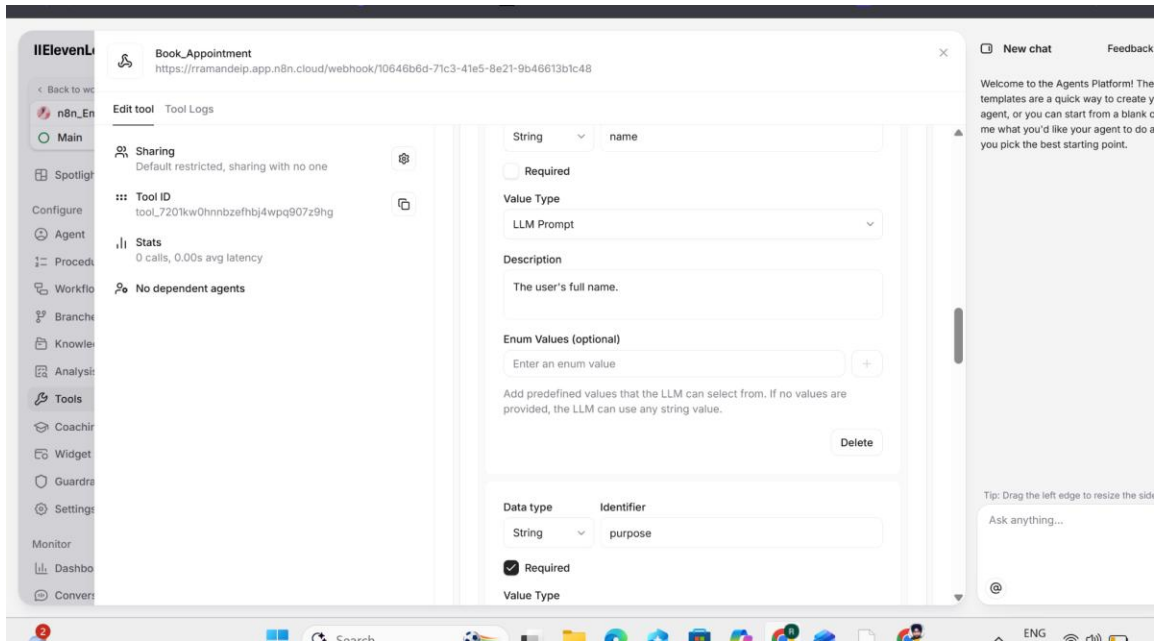
The screenshot shows the 'Edit tool' configuration page for a tool named 'Book\_Appointment'. The tool is configured with the following parameters:

- Name:** Book\_Appointment
- Description:** Use this tool whenever the user wants to book, schedule, reschedule, or confirm an appointment. Collect the following information from the user's voice request: Full Name
- Method:** POST
- URL:** <https://rramandeip.app.n8n.cloud/webhook/10646b6d-71c3-41e5-8e21-9b46613b1c48>
- Response timeout (seconds):** 20 seconds
- Disable interruptions:** ☐ (unchecked)

The page also includes a sidebar with navigation options like 'Main', 'Spotlight', 'Configure', 'Agent', 'Workflow', 'Branches', 'Knowledge', 'Analysis', 'Tools', 'Coach', 'Widget', 'Guardrails', 'Settings', 'Monitor', 'Dashboard', and 'Conversion'. A 'New chat' button is visible in the top right corner.

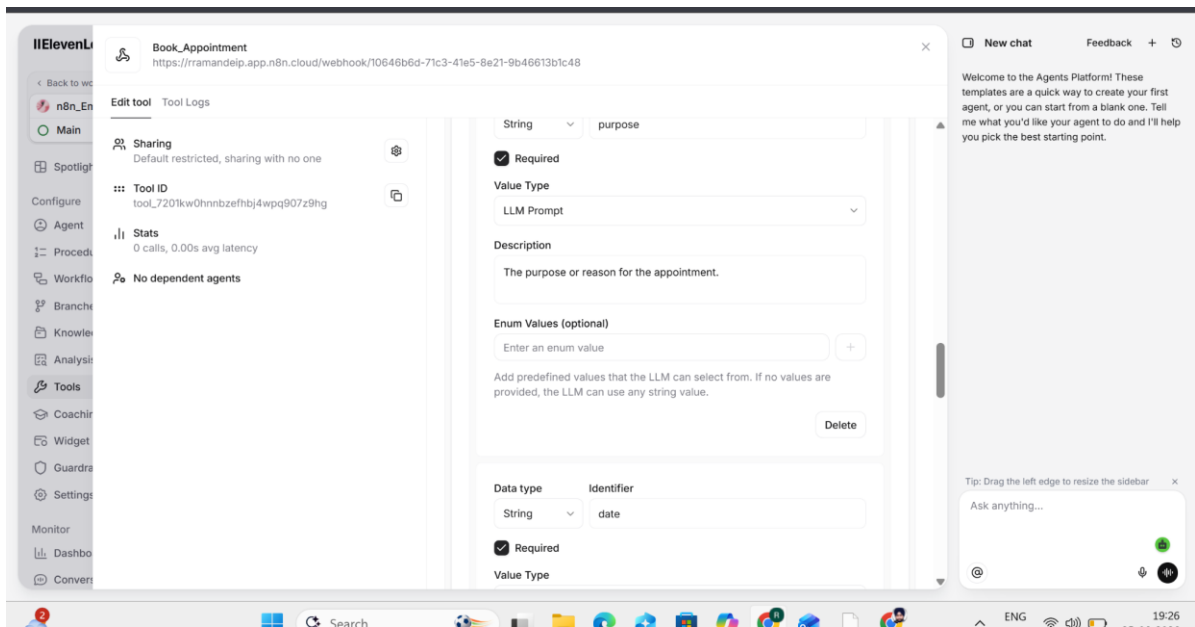
**Figure 5. Required Tool Parameters**

Definition of required tool parameter for user name.



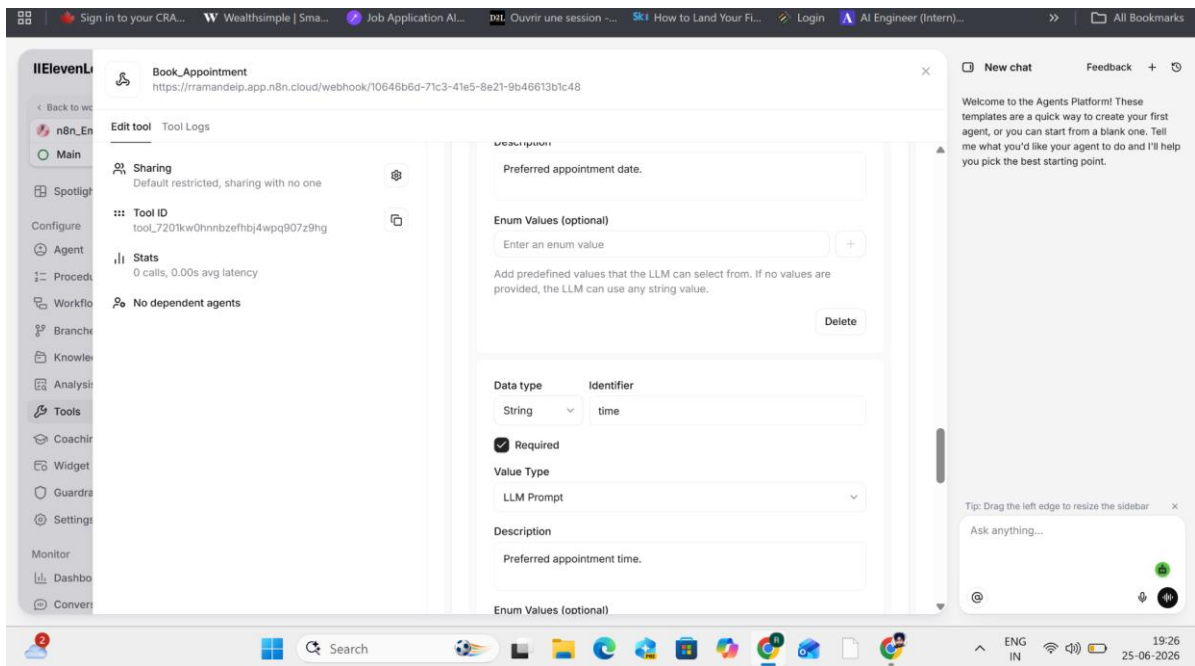
## Screenshot

Definition of appointment purpose parameter.



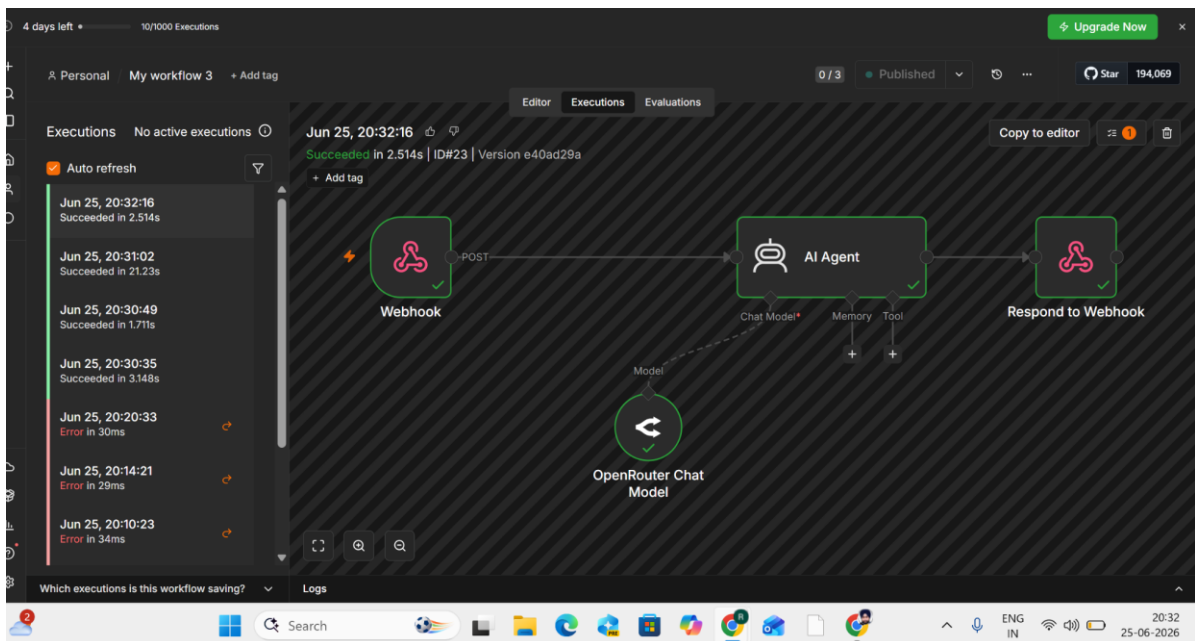
## Screenshot

Definition of appointment time parameter.



**Figure 6. Successful Workflow Execution**

Successful workflow execution history in n8n.



**Figure 7. AI-Generated Appointment Email**

Generated appointment email returned by the AI Agent.

AI Agent

INPUT

Schema

Table

JSON

Webhook

headers

T

host

rramandeip.app.n8n.cloud

T

user-agent

ElevenLabs/1.0

T

content-length

168

T

accept

/\*/\*

T

accept-encoding

gzip, br

T

cdn-loop

cloudflare; loops=1; subreqs=1

T

cf-connecting-ip

34.59.11.47

T

cf-ew-via

15

T

cf-ipcountry

US

T

cf-ray

a1181d238c2a7e6b-ORD

T

cf-visitor

{\"scheme\":\"https\"}

T

cf-worker

n8n.cloud

T

content-type

application/json

T

x-forwarded-for

34.59.11.47, 172.69.17.73

T

x-forwarded-host

rramandeip.app.n8n.cloud

T

x-forwarded-port

443

Parameters

Settings

Tip: Get a feel for agents with our quick [tutorial](#) or see an [example](#) of how this node works

Source for Prompt (User Message)  
Define below

Prompt (User Message)  
Name: {{ \$json.body.name }}  
Purpose: {{ \$json.body.purpose }}  
Date: {{ \$json.body.date }}

Name: Rramandeip Singh Purpose: AI Healthcare Lab Assig\_  
☐ Require Specific Output Format  
☐ Enable Fallback Model

Options

System Message  
You are a professional appointment booking assistant.

Chat Model \*  
Memory  
Tool

Output

Logs

Schema

Table

JSON

output

Subject: Appointment Request  
Dear Professor John,  
I hope you are doing well. My name is Rramandeip Singh. I would like to request an appointment regarding AI Healthcare Lab Assignment. My preferred date is July 10, 2026 at 2:00 PM. Please let me know if this time is convenient. If not, I would appreciate an alternative time.  
You can contact me.  
Thank you for your time and consideration.  
Kind regards,  
Rramandeip Singh